

CA Trader - Formula & Dashboard Specification

Reconciled model for confluence, pullback entry, stop, target and risk-reward validation.

Source basis: Uploaded CA Trader - Terminal2.pdf. The dashboard shows 40 factors, CE/PE outputs, and four separate mathematical popups. This report preserves that structure and makes the calculations internally auditable.

1. Executive audit of the current dashboard

Block	Displayed result	Audit
Entry	212.87	Displayed formula with 219.456, Delta 0.52, VWAP gap 42, RSI 58.4 and flow 0.58 yields about 214.19.
Stop	192.65	Displayed formula implies about 39.70 premium risk before any cap, not 20.22.
Target	253.31	Displayed multiplier would produce about 256.44; the shown result is exactly Entry + 2 x Risk.
R:R	1 : 2.0	Correct only if target is explicitly defined as a 2R target. Dynamic targets must display actual computed R:R.
Weights	135% total	Treat displayed weights as relative and normalize them to 100% before scoring.
Consensus	73.9% bullish	Keep as a calibrated presentation score unless statistically mapped to observed probability.

Core design principle: do not use one formula to simultaneously manufacture entry, stop and target. Keep direction, execution, risk and reward as separate model layers.

2. Recommended calculation engine

A. Normalized confluence score

$w_i^* = w_i / \text{SUM}(w_i)$; Score = $\text{SUM}(w_i^* \times s_i \times \text{Reliability}_i \times \text{Freshness}_i)$, followed by re-normalization of effective weights. Each signal s_i is bounded to [-1, +1].

B. Pullback entry

Estimate a pullback in option-price units from Delta x VWAP distance plus ATR_opt and a noise buffer. Reduce the required discount when directional confluence is strong, and cap the resulting entry inside live Bid/Ask bounds.

C. Stop / risk budget

Use the larger of structural invalidation converted through Delta and a volatility floor based on ATR_opt. Apply a bounded IV multiplier, add execution noise, then cap the final risk as a percentage of entry premium.

D. Target

Estimate underlying movement to Support/Resistance, cap it by ATR, and reprice the option using Delta/Gamma/Vega/Theta. Do not use 2R to manufacture the target; use 2R as the minimum reward test.

E. R:R gate

Risk = |Entry-Stop|. Reward = |Target-Entry|. R:R = Reward/Risk. A setup passes only when R:R >= configured minimum and all execution gates pass.

3. Compact production formulas

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w_i* = w_i / SUM(w_i)
Score = SUM(w_i* x s_i x Reliability_i x Freshness_i)
(re-normalize effective weights)

D_entry = clip(a_vwap x abs(Delta) x VWAPGap + a_atr x ATR_opt + NoiseBuffer, D_min, D_max)
Entry = CMP - D_entry x clip(1 - lambda x abs(Score), F_min, F_max)

StructuralRisk = abs(Delta) x D_invalidation
VolatilityRisk = k_atr x ATR_opt
RiskBeforeCap = max(StructuralRisk, VolatilityRisk) x VolMultiplier + NoiseBuffer
RiskBudget = min(RiskBeforeCap, MaxRiskPct x Entry)
CE Stop = Entry - RiskBudget
PE Stop = Entry + RiskBudget

D_target = min(distance_to_structure, k_target_atr x ATR_underlying)
ModelTarget = Entry + abs(Delta) x D_target + 0.5 x Gamma x D_target^2
+ Vega x ExpectedIVChange - abs(Theta) x ExpectedTimeElapsed

Risk = abs(Entry - Stop)
Reward = abs(ModelTarget - Entry)
RR = Reward / Risk
Trade allowed only if RR >= RR_min
```

4. Dashboard criteria map - all visible source factors

Factor	Model role	Formula layer
Underlying Spot Price	Price anchor	Entry / Stop / Target
Strike Price & Moneyness	Contract filter	Candidate selection
Time to Expiry	Time-value risk	Entry / Stop / Target
Delta	Spot-to-option translation	Entry / Stop / Target
Gamma	Convexity	Entry / Target
Theta	Time decay	Entry / Target
Vega	IV sensitivity	Entry / Stop / Target
Implied Volatility	Volatility regime	Entry / Stop
Volatility Skew & Surface	Relative IV context	Direction / Entry
Gamma Exposure (GEX)	Dealer positioning	Direction / regime
Order Flow Delta Volume	Immediate demand/supply	Direction / Entry
Market Liquidity & Spread	Execution quality	Hard gate / Entry
Option Trading Volume	Participation	Direction / confidence
Open Interest (OI) Walls	Structural positioning	Target / Direction
Change in OI	Position build-up	Direction
Put-Call Ratio	Cross-option positioning	Direction
Dealer Delta Hedging	Mechanical flows	Direction / Entry
Trend Momentum (ADX / MACD)	Trend strength	Direction / Entry
Intraday VWAP Benchmark	Fair-value anchor	Entry
Relative Volume (RVOL)	Participation expansion	Direction / Entry

Factor	Model role	Formula layer
Clustered Support & Resistance	Invalidation / objective	Stop / Target
RSI(14) Momentum & Divergence	Momentum state	Direction / Entry
ATR Realized Volatility	Move-size estimate	Stop / Target
Benchmark Index Dominance	Market regime	Direction
Sector Cluster Rotation	Cross-sectional regime	Direction
Global Markets & GIFT Cues	Overnight context	Direction
Currency Swings (USD/INR)	Macro sensitivity	Context / Direction
Commodity Prices (Crude/Gold)	Macro regime	Context / Direction
Economic Data (CPI / PMI)	Macro regime	Context
Central Bank Policy & Repo Rate	Macro regime	Context
Institutional Flows (FII/DII)	Capital-flow regime	Direction
Futures Basis & Roll Spread	Derivative positioning	Direction
Short Interest & Squeeze Ratio	Positioning risk	Direction / Risk
Consensus Earnings Estimates	Event context	Context
Earnings Beat / Miss Surprises	Catalyst	Direction / Target
Management Commentary & Capex	Forward guidance	Context
91D T-Bill & 10Y Sovereign Yield	Rate regime	Context
Dividend Yield & Ex-Dates	Contract adjustment	Target / event filter
Corporate Actions (Bonus/Splits)	Contract/event risk	Hard filter
CA AI Live News Intelligence	Event sentiment	Direction / Target

5. Recommended dashboard UI changes

Current display	Recommended display
135% Bullish	Normalized weight mass: 100%; Directional score: +x.xxx; Bullish display: xx.x%
Pullback Entry only	Show CMP, Bid, Ask/Mid, Pullback distance, Entry, and factor contribution
Stop Loss (Wider)	Show StructuralRisk, VolatilityRisk, RiskBeforeCap, RiskCap and Stop
Target	Show ModelTarget and 2R MinimumTarget separately
R:R 1:2.0	Show Actual R:R; separately show configured minimum R:R
Optimal Institutional Quality	Use descriptive execution/risk labels rather than quality rankings
Auto Trade	Keep as an action control separate from the mathematical score

6. Minimum data-quality and execution gates

- Freshness: reject or down-weight stale feeds.
- Bid/ask: use executable prices and block entries when spread exceeds configured threshold.
- Liquidity: require minimum volume/open interest and reasonable depth.
- Expiry: apply stricter rules as DTE approaches expiry.
- Event risk: corporate actions or major scheduled events can force a no-trade state.
- Missing inputs: do not silently substitute zeros; lower reliability or block the signal.
- Target feasibility: no trade when modeled target does not clear the required minimum R:R.

7. Backtesting / calibration checklist

Calibrate coefficients with walk-forward and out-of-sample tests. Include spread, slippage, brokerage, taxes and regime splits. Track win rate, average R, median R, profit factor, maximum drawdown, expectancy, slippage-adjusted expectancy, entry fill rate, stop-out rate and target-feasibility rate.

Note: This document is a mathematical/modeling specification based on the supplied dashboard screenshots/PDF. It does not establish that any parameter is optimal until validated on the strategy data.